

Machine Learning - Claims Forecasting

Project Summary:

This project aims to develop an ML pipeline to forecast the number of insurance claims per policy period (ClaimNb). Accurately predicting this is critical for insurance providers to optimize premium pricing, manage risk reserves, and identify potential high-risk customers.

Project Structure:

Phase 1: Loading and inspecting the data
Phase 2: Target value EDA (ClaimNb and Exposure)
Phase 3: Univariate EDA
Phase 4: Multivariate EDA
Phase 5: Poisson GLM
Phase 6: XGBoost
Phase 7: Model comparison and error analysis

EDA on the dataset freMTPL2freq

Phase 1: Loading and inspecting the data

Upon initial analysis, the following conclusions have been drawn:

1. There are a total of 678 013 samples, each having 12 features: 'IDpol', 'ClaimNb', 'Exposure', 'Area', 'VehPower', 'VehAge', 'DrivAge', 'BonusMalus', 'VehBrand', 'VehGas', 'Density', 'Region'. One feature is not meaningful (IDpol), and one is the target (ClaimNb) with Exposure used as an offset. There are no explicit null values.
2. Several conclusions drawn for numeric columns ('IDpol', 'ClaimNb', 'Exposure', 'VehPower', 'VehAge', 'DrivAge', 'BonusMalus', 'Density').
The vast majority of ClaimNb are valued at 0, several are valued at 1, and there are rare instances of other numbers up to the maximum of 16. There are no negative/invalid values.
Suspiciously low minimum value for Exposure (0.0027 ~ 1 day). No negative/invalid values, or unusual values ≥ 3 .
Expected values for VehPower.
Suspicious maximum for VehAge of 100, which needs further inspection (maybe adding an extra feature for invalid VehAge).
Suspicious max for DrivAge.
Expected range for BonusMalus (50 -350).
No abnormal values for Density.
3. Categorical columns ('Area', 'VehBrand', 'VehGas', 'Region') have a number of unique values ranging from 2 to 22.
4. Type for IDPol was float64, so I cast it to int64 (that makes more sense).

Phase 2: Target value EDA

Histogram for ClaimNb showed the vast majority of the dataset being valued at 0 claims, 0.05% valued at 1 claim, and 0.003% valued at 2 or more claims. The calculated **Dispersion Ratio** for the target is 1.08, which is remarkably close to the theoretical Poisson ideal of 1.0. This indicates that the mean and variance are nearly equal, which suggests that a Poisson GLM is an appropriate choice for this dataset. However, XGBoost will still be used to move beyond the linear assumptions of the GLM. There might be non-linear interactions between variables which a tree-based model could uncover.

Data profiling of the **Exposure** variable revealed a minimum value of 0.00272 (~ 1 day). I implemented an Exposure floor at the level 0.003 to serve as a safety constraint. This ensures that even if future data contains negative or near-zero exposures, the log-offset used in the Poisson model remains numerically stable and prevents unrealistic frequency spikes from distorting the model coefficients.

It was noticed that although extremely small exposures account for only 0.2844% of all exposures, they also account for 3.27% of all claims. This could be due to noise (division by an extremely small number) and will be accounted for in EDA.

Frequency distribution revealed a heavy right-tail with extreme values exceeding 600 claims/year, which are considered **outliers**. They typically represent policies with very short durations (~ 1 day) where a claim occurred. While mathematically valid, they don't represent a predictable risk trend. In the modeling phase, their influence was mitigated by using a Poisson log-offset, which naturally weights observations by their exposure, ensuring that a one day policy doesn't carry the same predictive weight as a one year policy.

Claim rate by exposure bins: The high claim rate in the (0–0.05] exposure bin is likely driven by the extreme outliers (more than 300 claims/year). They look risky, but there are very few of them, so they actually represent noise. The (0.5–1.0] bin represents the standard annual policies. While their rate looks lower on the bar chart, they represent the bulk of the company's exposure and total loss, and they are the actual insurance risk I want the model to learn. That's also a reason behind choosing Poisson GLM with Exposure as an offset.

Phase 3: Univariate EDA

For each feature, I plotted the histogram, as well as a combined plot of the exposure volume and risk (claim frequency). By including the global average frequency in the plot, I made it easier to determine which customers are above/below the average risk. It is also easier to decipher whether a spike in risk rate is due to real threats or simply noise due to a decline in the number of samples (meaning that there are not enough samples to make a knowledgeable decision).

VehAge valued up to one year proved to have the highest risk for claiming insurance, which was expected due to a large part of overall vehicles being up to a year old, as well as the fact that drivers of new cars are more likely to claim insurance than those owning older vehicles. However, VehAge of 99 and 100 showed spikes with 23 and 25 samples respectively, and were therefore labeled as NaN and will be imputed as median in model pipeline.

Histogram for **DrivAge** showed a normal distribution. The risk was highest for young drivers, which is to be expected, gradually lowering itself until a slight rise in the ages 44 - 53. DrivAge of 99 showed a spike of 70 samples, was therefore labeled as NaN and will be imputed as median in the model pipeline.

The following **business-related conclusions** have been drawn:

- **VehAge**: Implement a higher premium for new cars because the likelihood to claim is statistically much higher (claiming for minor damage because the car is new). For vehicles in the 10+ year bracket, the company should urge the customers to purchase lower-coverage products since the repair of a total damage might exceed the cost of the car. Marketing should focus on cars in the 3–8 year range, because these show the best balance of high exposure and below-average risk.
- **DrivAge**: Might be a good idea to implement tier-pricing a.k.a. lowering prices proportionally to years of driving experience. Also monitor high-risk behavior (young drivers) in real-time and offer lower rates only to those who prove they are safe. For drivers over age 70, the company should move toward a more manual review process, because the data is sparse (low exposure).
- **BonusMalus**: Customers valued at 50 are the lowest-risk and highest-volume customers, so for them there should be a loyalty reward or price-matching for competitor insurance companies. The risk for drivers in the 95–230 range is so high (0.30+ frequency) that standard premiums will not cover their expected losses. They require a higher surcharge. Increase in the 60-64 bin is for drivers who have had their first strike. They could be given a pass or a slight increase in price, depending on company's decision.
- **Density**: There is a nearly perfect upward trend for claim frequency as population density rises. Exposure volume is well-distributed across all density levels, meaning the rising risk trend is statistically robust and not caused by noisy small exposure values. Suggestion is territorial pricing: drivers in the highest density areas should have a surcharge, whereas those in lower density areas should be offered discounts.
- **VehPower**: For vehicles with a power rating of 9 or higher should face higher prices, since the data shows they are significantly riskier than the average. The 7-8 power range has decent exposure and the lowest observed frequency, making them highly profitable customers. It's intriguing that low-power cars (<5) have higher risk than mid-power ones. This might be because they are driven by young drivers in cities, and if that's the case, I will check in multivariate EDA if the DrivAge and Area factors are correctly capturing that risk to avoid over-pricing the vehicle itself.
- **Area**: Areas **F** and **E** should have significant increase in premium prices to account for the high claim frequency, whereas areas **A** and **B** have risk which is significantly below the global average, meaning that we can offer lower premiums to attract these customers. They are highly profitable because their likelihood of claiming is much lower.
- **VehBrand**: data reveals that **B1** and **B2** represent the most stable components of the dataset. Despite accounting for a major share of total exposure, their observed claim frequency remains consistently below the global average. This suggests these brands attract a lower-risk profile of drivers, representing a highly profitable segment where the company should focus its retention efforts. On the contrary, even though **B12** is popular among drivers, they crash too much, maybe warranting for higher rates.
- **VehGas**: while the data shows a slight difference in observed frequency between Diesel and Regular vehicles, the variance is marginal (<1%). Both segments possess

massive exposure volume, confirming that this near-equality is a stable characteristic of the portfolio. From a pricing perspective, this variable lacks materiality. I recommend no pricing action for VehGas.

- Region: possible price increase for **R82** and **R11** since they have significant exposure (meaning that this isn't just noise, but a consistent pattern of high claims that needs to be offset by higher premiums). Might be a good idea to win-over customers in **R23** and **R53**. These are underrated regions, the data shows they are safer than average, so they might be more profitable than a policy written in the big cities.

Phase 4: Multivariate EDA + Translating Multivariate EDA into Model Predictors

- BonusMalus vs DriverAge: As we move from the bottom to the top row, the risk changes dramatically from **0.057** to **0.231**, meaning that no matter how old a driver is, their historical malus is a massive warning sign of future claims. While risk does generally increase as we move from left to right in the middle and top rows, the change is less extreme than the vertical change.
- Area vs DrivAge: A young driver is not automatically the most dangerous person; it depends entirely on where they drive. An older driver in area F is riskier than a teenager in area A.
- Area vs Density: The heatmap looks like a diagonal staircase with lots of empty white spaces, meaning that these two variables are **highly collinear**. Therefore, I won't use both variables in my final model, in order to prevent overfitting. I will keep Density because it is a continuous numeric feature providing more information, rather than a sparse categorical one such as Area.
- Area vs Region: In R94 and R11, the risk is high even in the rural areas (Area A and B). For example, a driver in Area B in R24 has a risk of 0.081, but a driver in Area B in R94 has a risk of 0.176. This means Region is not redundant. There is something about the geography or local driving culture in R11 and R94 that makes them twice as dangerous as R24, regardless of whether you are in a city or a village.
- Region vs Density: Unlike the *Area vs Density* heatmap which was diagonal, this grid is much fuller. Most regions, like R24, have values across almost every density level, from the lowest (0.079) to the highest (0.128). This means they are separate variables that are not colinear. Also, if risk was only about density, every region in the same density column would have the same color, which isn't the case.
- VehPower vs DrivAge: The young drivers show a dangerous climb in risk as vehicle power increases. A low-power car for a young driver has a frequency of 0.120, but a high-power car (8.0–15.0) spikes the risk to 0.146, which is the highest in the heatmap. On the other hand, for more experienced drivers, vehicle power barely matters. Their risk ranges between 0.092 and 0.104.

As I previously mentioned in univariate EDA, some drivers might be overpriced just for the power of the car itself. That turned out to be the case here for experienced drivers (40+ years), meaning that they could be offered better rates while also maintaining high profitability. Therefore, I included an interaction feature for *DrivAge* and *VehPower* in the form of a flag 'Is_New_Veh'.

As for the significance of *VehAge*, I added a new feature 'Is_New_Veh' for vehicles under 2 years, since the risk is mostly concentrated in brand-new vehicles.

Pearson and Spearman correlation:

Because I discovered a near-perfect collinearity between Area and Density, I decided to include a correlation analysis with both Pearson and Spearman methods. The primary objective was to identify any remaining redundant features that could destabilize the Poisson GLM or lead to inflated standard errors.

I used two different types of correlation to capture different types of relationships. While Pearson correlation measures linear relationships between features, Spearman measures monotonic relationships, which can sometimes be more telling if the relationship between features isn't a straight line.

No other pairs showed a correlation above 0.6, so the remaining features are sufficiently independent to be included in the model together.

As for those with the highest correlation factors, Pearson of -0.45 and Spearman of -0.41 showed that higher-risk brands might be associated with newer vehicles, which again proves why I need the 'Is_New_Veh' flag (it ensures I don't just blame the brand for the high risk, when the actual cause might be that the driver is new to this car).

Phase 5: Poisson GLM

I chose the Poisson GLM because the target variable (ClaimNb), consists of non-negative integers (counts). Unlike standard linear regression, Poisson regression is specifically designed to handle the long-tail nature of insurance claims where most values are zero. As was previously mentioned in Phase 3: the Dispersion Ratio for the target is 1.08, which is remarkably close to the theoretical Poisson ideal of 1.0, suggesting that a Poisson GLM is an appropriate choice.

What is more, the Poisson GLM allows for the direct integration of Exposure as a logarithmic offset. This ensures we are modeling the claim rate (frequency) while maintaining the integrity of the original count distribution, rather than simply dividing the target and losing statistical nuance.

For the target I chose ClaimNb, with Exposure as an offset.

$$y = \text{Features} * \text{Weights}$$

$$\frac{\text{Claims}}{\text{Exposure}} = e^{\text{Features} * \text{Weights}}$$

$$\log(\text{Claims}) = \log(\text{Exposure}) + \text{Features} * \text{Weights}$$

For **VehAge** and **DrivAge**, NaN values were handled by imputing them as median.

To better align with the log-linear assumptions of the Poisson GLM, the *Density* feature was log-transformed, because it had high right-skewness. The original Density feature was dropped, due to its colinearity to *Log_Density*.

By introducing a new feature *Is_New_Veh*, i made sure that the model treats new cars as a special category, as the earlier analysis showed that vehicles aged 0-2 years have a much higher claim frequency than older cars.

Young_HighPower is an interaction eature between DrivAge and VehPower. The Multivariate EDA showed that a young driver in a fast car is a unique danger zone (risk spike of 0.146). This flag allows the model to add an extra penalty to this specific combination.

I did **one-hot encoding** for categorical features ('VehBrand', 'Region', 'VehGas'). I dropped the 'Area' feature because of the high colinearity to the 'Density'.

The data was split into training and test sets in the **80/20 ratio**. Before training, 'ClaimNb', 'Exposure' and 'IDPol' were dropped. **StandardScaler** was used on numeric columns. I did a **grid search with k-fold** cross-validation (k=5) in search for the best hyper-parameters. The maximum number of iterations was 2000.

In the end, I reported the evaluation metrics **MAE, RMSE, Mean Deviance and Total Deviance** both on the training and the test set.

The Stability Check (Train vs. Test)

There is a consistency between the training and test sets, which means there isn't overfitting:

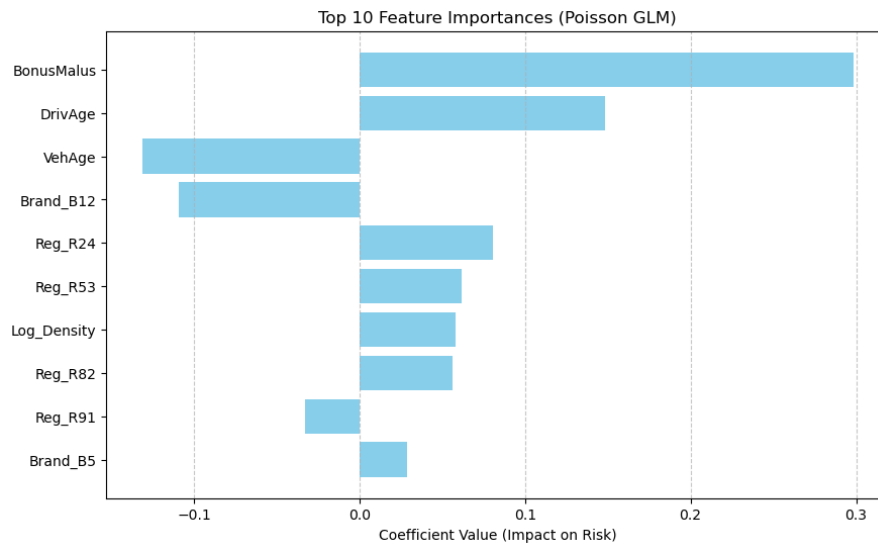
Dataset	MAE	RMSE	Mean Deviance	Total Deviance
Full Training Set	0.083193	0.240857	0.340071	184457.76
Test Set	0.083302	0.238609	0.340704	46200.48

Mean Deviance of 0.3407 indicates that the model is significantly more accurate than a simple average-only model. MAE (0.0833) shows that, on average, the model's prediction is only off by 0.08 claims per policy. Since most drivers have 0 claims, this shows the model is correctly assigning very low probabilities to the majority of the portfolio.

Best Alpha (0.01): The grid search chose a small regularization value. This confirms that the model didn't need much automated help to stay stable and not overfit.

The comparison between the actual data (*y_train*) distribution and the model predictions highlights the expected behavior of a Poisson GLM. The raw data contains extreme outliers up to 16 claims, but the model predicts a narrower range of expected frequencies (0 to 3) and successfully captures the mean risk of the dataset. This means that model doesn't react to individual random noise.

These are the top 10 most informative features:



Phase 6: XGBoost

In order to capture complex, non-linear interactions within the freMTPL2freq dataset, I implemented an XGBoost. Unlike the Poisson GLM, there was no need for one-hot encoding for categorical features, since XGBoost has a native categorical support. Other than that and not using StandardScaler, all of data pre-processing was done the same way.

To find the best hyper-parameters, I used Bayesian Search with k-fold cross validation.

Phase 7: Model comparison and error analysis

These are the metrics I got for both models:

Metric	Poisson GLM	XGBoost
Mean Poisson Deviance	0.3329	0.3197
Total Deviance	45148	43353
MAE	0.0827	0.0812
RMSE	0.2376	0.2357

The XGBoost model achieved a Mean Deviance of 0.3197, a significant improvement over the GLM's 0.3329. XGBoost better captured the underlying frequency distribution, because it can model non-linear interactions between features like Age, Region, and BonusMalus that the GLM might miss.

--- Top 10 XGBoost Residuals ---						
	Actual_Claims	Predicted_Count	Exposure	Residual	DrivAge	VehBrand \
304484	6	0.067344	0.33	5.932656	52.0	B1
364437	4	0.024721	0.27	3.975279	23.0	B3
4808	3	0.007393	0.09	2.992607	28.0	B12
497569	3	0.012053	0.21	2.987947	53.0	B12
214606	3	0.022375	0.32	2.977625	24.0	B1
535261	3	0.040455	0.77	2.959545	62.0	B2
7154	3	0.042573	0.65	2.957427	41.0	B1
185528	3	0.054807	0.97	2.945193	35.0	B1
630041	3	0.059721	0.91	2.940279	52.0	B2
510386	3	0.078448	0.67	2.921552	34.0	B13

	Region	BonusMalus
304484	R91	50
364437	R11	90
4808	R11	100
497569	R11	52
214606	R94	90
535261	R31	60
7154	R24	50
185528	R93	57
630041	R94	50
510386	R31	78

Most of the errors seen in the residuals tables are the same customers, which means that the models are consistently failing on the same outliers. Both models have their largest error on the driver who had 6 claims in just 0.33 years of exposure. This driver is 52, has a perfect BonusMalus of 50, and lives in Region R91. No demographic features can justify this statistically, so I consider it as an extreme behavioral outlier.

I noticed that many IDs appeared in both lists (e.g. 304484, 364437, 497569, 535261, 7154, 185528, 630041). Since a complex non-linear model (XGBoost) and a simple linear model (GLM) both failed similarly, I would say that these drivers were simply unpredictable based on the available features.

Also, some of the largest residuals have low Exposure (e.g., ID 497569 with 0.21, ID 304484 with 0.33). I already mentioned that Poisson GLM treats Exposure as a weight, so a claim occurring in a very short policy period creates a massive frequency that the model struggles to predict considering the driver's stable demographics.

Suggesting improvements:

- **Maybe trying the target as $\log(1 + \text{claims/exposure})$ or something similar. By doing this, I got the mean poisson variance of ~ 0.16 , however I didn't know if such a transformation of target feature would be acceptable, so I left it as is.**

- Treating it as a classification problem, where for example one class would be 0 claims, second class 1 claim, third class 2+ claims (imbalanced class problem).

- Implementing a two-stage (hurdle) model: Instead of one model trying to do everything, we could have a binary classifier (XGBoost Classifier) to predict if a person is likely to have at least one claim, and then have a Poisson regressor to predict how many claims they had if the answer to the first question was yes.

- Dynamic Weighting (DART Booster): In the XGBoost, maybe using the DART (Dropouts meet Multiple Additive Regression Trees) booster instead of the standard gbtrees. Because standard boosting can over-correct for outliers like the 6-claim driver, leading to the near-zero predictions we are seeing for everyone else. DART randomly drops trees during the

training process, which prevents any single outlier from having too much influence on the final model's weights.

- Additional features: Since the residual analysis showed that drivers with "safe" profiles (Age 52, perfect BonusMalus) are still having multiple claims, the model clearly lacks more behavioural data. Maybe adding external data and features which could explain the random accidents that Age and Region can't account for (e.g. Average Speeding Over Limit, Annual Mileage, Number of Drivers on Policy, Vehicle Safety Rating).